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# PART 1: SYNTHETIC DATASET (np.polyfit)

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import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import r2_score

print("==== PART 1: SYNTHETIC DATASET =====\n")

# Create dataset
x = np.array([95, 85, 80, 70, 60])
y = np.array([85, 95, 70, 65, 70])

# Linear regression using polyfit
model = np.polyfit(x, y, 1)

print("Model Coefficients (slope, intercept):", model)

# Prediction
predict = np.poly1d(model)
print("Prediction for x = 65:", predict(65))

# Predict all values
y_pred = predict(x)
print("Predicted values:", y_pred)

# R2 Score
print("R2 Score:", r2_score(y, y_pred))

# Plot

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y_line = model[1] + model[0] * x
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plt.figure()
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```
plt.scatter(x, y, color='blue', label="Actual")
```

```
plt.plot(x, y_line, color='red', label="Regression Line")
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```
plt.scatter(x, y_pred, color='green', label="Predicted")
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```
plt.xlabel("X")
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```
plt.ylabel("Y")
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```
plt.title("Synthetic Data Linear Regression")
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plt.legend()
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plt.show()
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# PART 2: BOSTON HOUSING DATASET
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import pandas as pd
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```
from sklearn.model_selection import train_test_split
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from sklearn.linear_model import LinearRegression
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from sklearn.metrics import mean_squared_error
```

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print("\n==== PART 2: BOSTON HOUSING DATASET =====\n")
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# Load dataset (change path if needed)
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df = pd.read_csv("/content/Boston.csv")
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print("Columns in dataset:")
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```
print(df.columns)
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# Remove unwanted column if exists
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if 'Unnamed: 0' in df.columns:

    df.drop(['Unnamed: 0'], axis=1, inplace=True)

# Handle missing values by dropping rows with NaN
df.dropna(inplace=True)

# Check missing values after dropping NaNs
print("\nMissing Values After Dropping Rows:")
print(df.isnull().sum())

# Dataset overview
print("\nShape:", df.shape)
print("\nStatistical Summary:\n", df.describe())

# =====
# AUTO-DETECT TARGET COLUMN (NO ERROR)
# =====

target_col = None
for col in df.columns:

    if col.lower() in ['medv', 'price']:

        target_col = col

if target_col is None:

    raise Exception("Target column not found! Check dataset column names.")

print("\nTarget Column Detected:", target_col)

# Split features and target
X = df.drop(target_col, axis=1)
y = df[target_col]

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# Train-test split
xtrain, xtest, ytrain, ytest = train_test_split(X, y, test_size=0.2, random_state=0)

# Train model
model = LinearRegression()
model.fit(xtrain, ytrain)

# Predictions
ytrain_pred = model.predict(xtrain)
ytest_pred = model.predict(xtest)

# Evaluation
print("\nTrain MSE:", mean_squared_error(ytrain, ytrain_pred))
print("Test MSE:", mean_squared_error(ytest, ytest_pred))

print("\nTrain R²:", r2_score(ytrain, ytrain_pred))
print("Test R²:", r2_score(ytest, ytest_pred))

# Plot results
plt.figure()
plt.scatter(ytrain, ytrain_pred, label="Train Data")
plt.scatter(ytest, ytest_pred, label="Test Data")

plt.xlabel("True Values")
plt.ylabel("Predicted Values")
plt.title("True vs Predicted (Boston Dataset)")
plt.legend()
plt.show()
```